

HKCEE 1986
Mathematics II

Label 1 If $r = \sqrt[3]{h^3 - 7r^3}$, then the ratio $r : h$ is

- A. 1 : 8
- B. $1 : 2\sqrt{2}$
- C. $1 : 2$
- D. $\sqrt{2}$
- E. $1 : \sqrt[3]{2}$

Label 2 $\frac{x^2}{3} - 3y^2 = \frac{3}{2}(x - 3y)$

- A. $\frac{1}{2}(x - 3y)$
- B. $\frac{2}{9}(x - 3y)$
- C. $2(x + 3y)$
- D. $\frac{1}{2}(x + 3y)$
- E. $\frac{2}{9}(x + 3y)$

Label 3 If $1 - \frac{x+y}{y-x} = a$ ($a \neq 0$), then $y =$

- A. a
- B. $\frac{x}{a}(a - 2)$
- C. $\frac{x}{a}(a - 1)$
- D. $\frac{x}{a}(2 - a)$
- E. $\frac{x}{a}(1 - a)$

Label 4 Which of the following is an identity/are identities?

- I. $(x + 1)(x - 1) = x^2 + 1$
- II. $x^2 - 2x + 1 = 0$

Label 5 $(x - 2)^2 = (2 - x)^2$

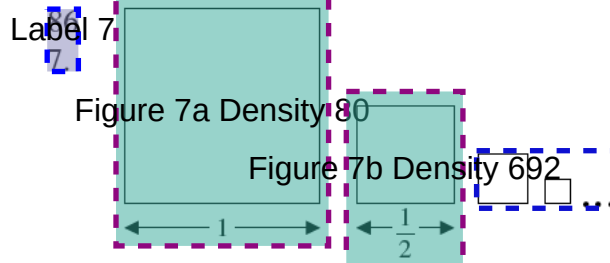
- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I and III only

Label 5 Given the identity $\frac{2}{x-1} + \frac{x+1}{(x-1)^2} + \frac{a}{(1-x)^2} = \frac{bx-2}{(x-1)^2}$ find the values of the constants a and b

- A. $a = 1, b = 3$
- B. $a = 3, b = 1$
- C. $a = 1, b = -3$
- D. $a = 3, b = -1$
- E. $a = -1, b = 3$

Label 6 If α and β are the roots of the equation $2x^2 + x + 3 = 0$, find the value of $\alpha - \frac{\alpha^2}{\alpha + \beta}$

- A. -3
- B. -2
- C. 2
- D. 3
- E. It cannot be determined

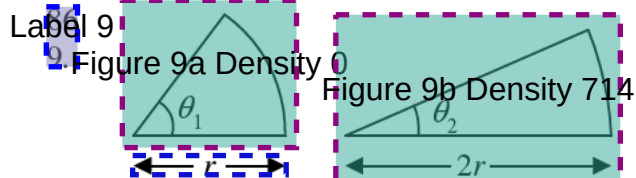


The figure shows an infinite number of squares. The length of a side of the first square is 1. The side of the first square is equal to half of the side of the preceding one. Find the sum of the areas of the infinite number of squares.

- A. 4
- B. 2
- C. $\frac{5}{3}$
- D. $\frac{3}{2}$
- E. $\frac{4}{3}$

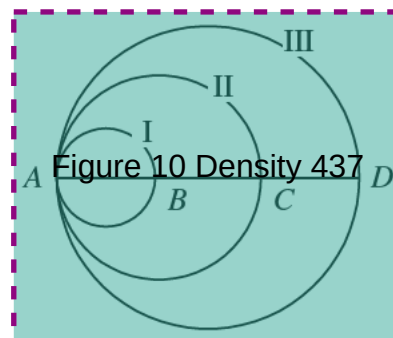
Find the real value of x such that $\begin{cases} x^2 + x + 1 = k \\ x - 1 = \frac{7}{k} \end{cases}$, where k is a constant.

- A. 3
- B. 2
- C. 1
- D. -1
- E. -3



The figure shows two sectors with radii r and $2r$. If these two sectors are equal in area, then $\theta_1 : \theta_2 =$

- A. $2 : 1$
- B. $3 : 1$
- C. $4 : 1$
- D. $5 : 1$
- E. $6 : 1$

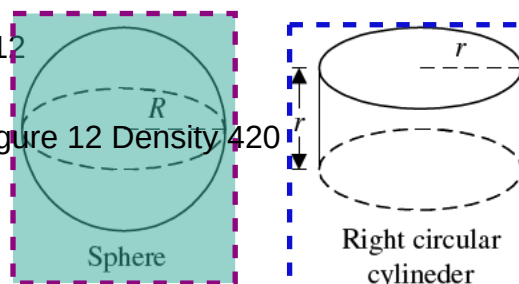


In the figure, $ABCD$ is a straight line, with $AB = BC = CD$. Three circles I, II and III are drawn respectively on AB , AC and AD as diameters. Areas of circle I, Area of circle II, Area of circle III =

- A. $1 : 2 : 3$
- B. $1 : 2 : 4$
- C. $1 : 4 : 9$
- D. $1 : 4 : 16$
- E. $1 : 8 : 27$

A man drives a car at 45 km/h for 3 hours and then at 50 km/h for 2 hours. His average speed for the whole journey is

- A. 47 km/h
- B. 47.5 km/h
- C. 48 km/h
- D. 48.5 km/h
- E. 49 km/h



In the figure, if $\frac{\text{Volume of the sphere}}{\text{Volume of the right circular cylinder}} = \frac{9}{2}$, then $\frac{R}{r} =$

- A. $\frac{3}{2}$
 B. $\frac{3}{\sqrt{2}}$
 C. $\frac{3}{\sqrt{2}}$
 D. $\frac{3}{\sqrt{2}}$
 E. $\frac{9}{2}$

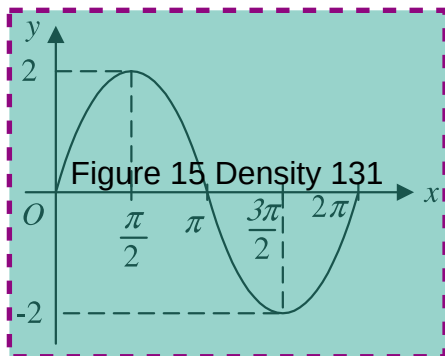
Label 13
 13 If the compound interest on \$1000 for two years at 9% p.a., pay half-yearly is \$x, find x.

- A. $1000 \times \frac{9}{100} \times 2$
 B. $1000(1 + \frac{9}{100})^4$
 C. $1000(1 + \frac{4.5}{100})^4$
 D. $1000(1 + \frac{9}{100})^2 - 1000$
 E. $1000(1 + \frac{4.5}{100})^4 - 1000$

Label 14
 14 If $\sin \theta \cos \theta = \frac{1}{4}$, then $(\sin \theta + \cos \theta)^2$

- A. $\frac{2}{3}$
 B. $\frac{3}{2}$
 C. $\frac{1}{2}$
 D. $\frac{1}{2}$
 E. $\frac{1}{4}$

Label 15
 15



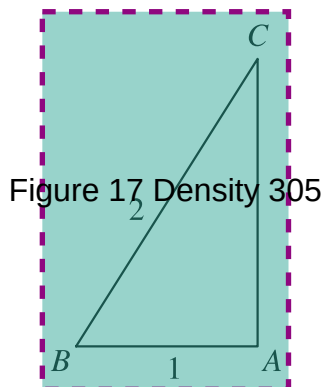
Which of the following functions may be represented by the above graph in the interval 0 to 2π ?

- A. $y = \cos 2x$
 B. $y = 2 \cos x$
 C. $y = \frac{1}{2} \cos 2x$
 D. $y = \sin 2x$
 E. $y = 2 \sin 2x$

Label 16
 16 $\sin^4 \theta - \cos^4 \theta =$

- A. -1
 B. $1 - 2 \cos^4 \theta$
 C. $\sin \theta - \cos \theta$
 D. $\sin^2 \theta - \cos^2 \theta$
 E. $2 \sin^4 \theta - 1$

Label 17
 17



In the figure, $\angle A : \angle B : \angle C =$

- A. $2 : \sqrt{3} : 1$
 B. $4 : 3 : 1$
 C. $3 : 2 : 1$
 D. $\sqrt{3} : \sqrt{2} : 1$

E. $1 : 2 : \sqrt{3}$

Label 18

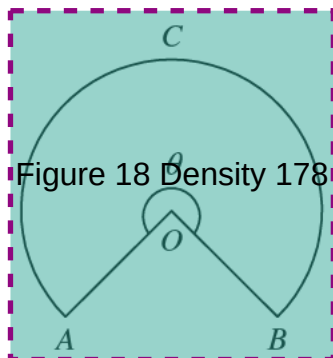


Figure 18 Density 178

In the figure, if the area of the sector is x , then arc $ACB =$

- A. $\frac{2x}{r}$
- B. $\frac{x}{r}$
- C. $\frac{2x}{r^2}$
- D. $\frac{\pi x}{90r}$
- E. $\frac{90x}{\pi}$

Label 19

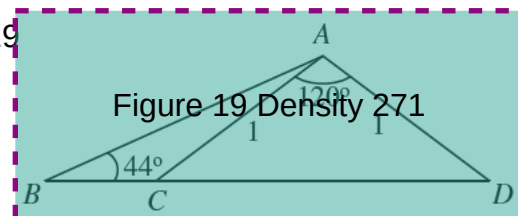


Figure 19 Density 271

In the figure, $AC = AD = 1$, $\angle ABD = 20^\circ$ and $\angle CAD = 120^\circ$, find AB .

- A. $2 \cos 20^\circ$
- B. $\frac{1}{2 \sin 20^\circ}$
- C. $\frac{\sqrt{3}}{2 \sin 20^\circ}$
- D. $\sqrt{3} \cos 20^\circ$
- E. $2 \sin 20^\circ$

Label 20 The bearing of a lighthouse as observed from an ocean liner is $N37^\circ E$, the bearing of the ocean liner as observed from the light house is

- A. $N37^\circ E$.
- B. $N53^\circ W$.
- C. $S37^\circ E$.
- D. $S37^\circ W$.
- E. $S53^\circ W$.

Label 21 Which of the following represents a circle?

- A. $2x^2 - 8y + 5 = 0$.
- B. $2x^2 + y^2 - 4x - 3y = 0$.
- C. $3x^2 + 3y^2 - 5x - 7 = 0$.
- D. $x^2 - y^2 - 7x + 6y + 1 = 0$.
- E. $x^2 + y^2 + 2xy + 7y - 1 = 0$.

Label 22

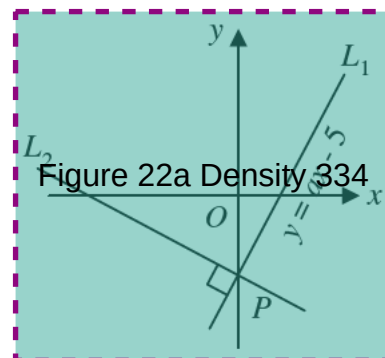


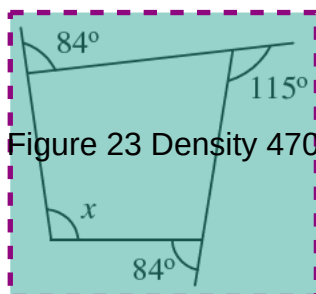
Figure 22a Density 334

In the figure, L_1 and L_2 are two straight lines perpendicular to each other and intersecting at P on the y -axis. If the equation of L_1 is $y = ax - 5$, then the equation of L_2 is

- A. $y = -\frac{1}{a}x - 5$
- B. $y = -\frac{1}{a} + 5$
- C. $y = -ax - 5$
- D. $y = -\frac{1}{a}x + 5$
- E. $y = -\frac{1}{a}x$

Figure 22b Density 1161

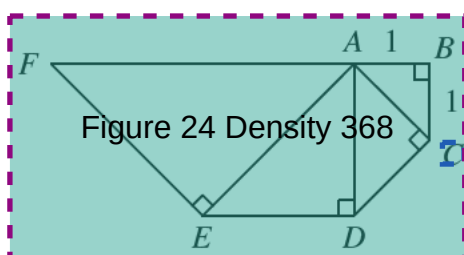
Label 23



In the figure, $x =$

- A. 77°
- B. 84°
- C. 96°
- D. 103°
- E. 115°

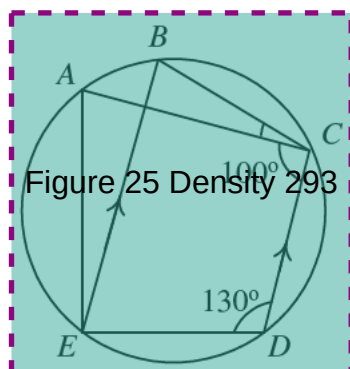
Label 24



In the figure, ABC , ACD , ADE and AEF are right angled isosceles triangles. If $AB = BC = 1$, how long is AF ?

- A. $2\sqrt{5}$
- B. 4
- C. $2\sqrt{3}$
- D. 3
- E. $\sqrt{5}$

Label 25



In the figure, A , B , C , D and E lie on a circle. AC intersects BE at K . $\angle ACD = 100^\circ$ and $\angle CDE = 130^\circ$. If $BE \parallel CD$, then $\angle ACB =$

- A. 25°
- B. 30°
- C. 36°
- D. 40°
- E. 42°

Label 26 In a shooting game, the probabilities for

John and Mary to hit a target are $\frac{4}{5}$ and $\frac{3}{5}$ respectively. When both shoot at the target, what is the probability that they both miss.

- A. $\frac{2}{25}$
- B. $\frac{3}{25}$
- C. $\frac{8}{25}$
- D. $\frac{12}{25}$
- E. $\frac{13}{25}$

Label 27 Given two groups of numbers

$$a + 1, a + 2, a + 3 \text{ and } b + 1, b + 2, b + 3,$$

where $a > b$. m_1 and m_2 are respectively the means of the two groups, and s_1 and s_2 are respectively their standard deviations. Which of the following is true?

- A. $m_1 > m_2$ and $s_1 > s_2$
- B. $m_1 > m_2$ and $s_1 = s_2$
- C. $m_1 = m_2$ and $s_1 > s_2$
- D. $m_1 = m_2$ and $s_1 = s_2$
- E. $m_1 > m_2$ and $s_1 < s_2$

Label 28
28

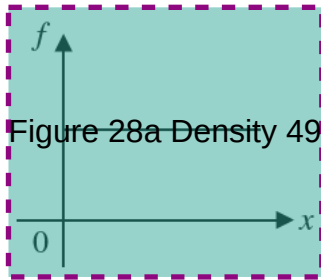


Figure 28a Density 49

The figure shows the frequency curve of a certain distribution. Which of the following can be the distribution's cumulative frequency curve?

A

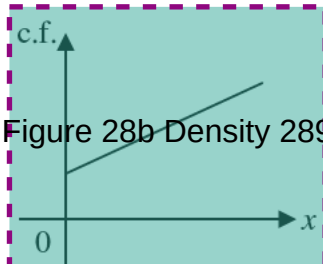


Figure 28b Density 289

B

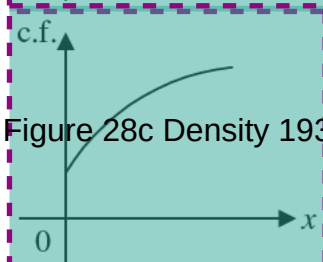


Figure 28c Density 193

C

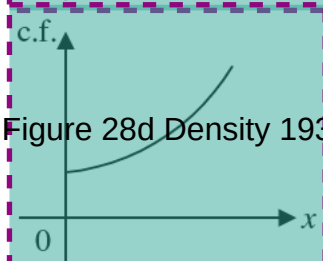


Figure 28d Density 193

D

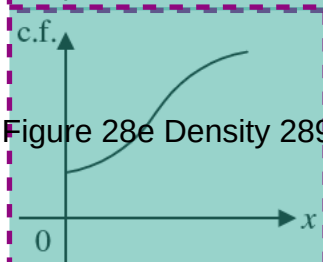


Figure 28e Density 289

E

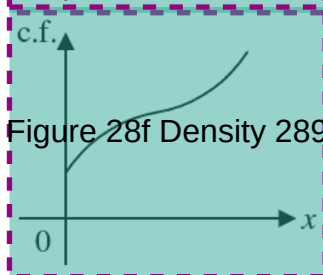


Figure 28f Density 289

Label 29
29 If $(10^x)^y = (2^z)(5^z)$, then which of the following must be true

A

$xy = z$

B

$xy = 2z$

C

$xy = z^2$

D

$x^y = z$

E

$x^y = 2z$

Label 30
30 Figure 30 Density 0

$\left(\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}}\right)^2$

A

$\frac{(x+y)^2}{xy}$

B

$\frac{x^2+y^2}{xy}$

C

$\frac{x+y+2}{xy}$

D

$\frac{x+y}{xy}$

E

$\frac{xy}{x+y}$

Label 31
31 The L.C.M. of $12a^2b$ and $18ab^3c$ is

A

$6ab$

B

$6a^2b^3c$

C

$36ab$

D

$36a^2b^3c$

E

$216a^3b^4c$

Label 32
32

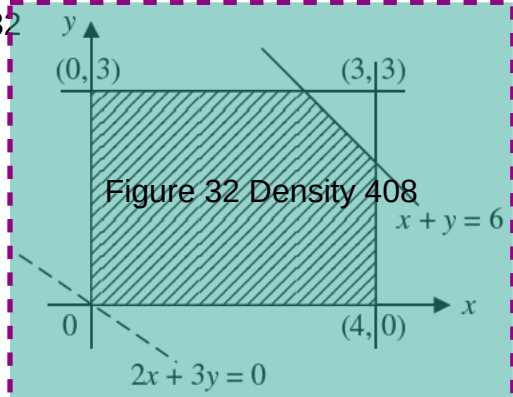


Figure 32 Density 408

Let $p = 2x + 3y$. Under the following constraints

$$\begin{cases} x \geq 0 \\ y \geq 0 \\ x \leq 4 \\ y \leq 3 \\ x + y \leq 6 \end{cases}$$

what is the greatest value of p ?

- A. 8
- B. 14
- C. 15
- D. 16
- E. 17

Label 33 If $\log x^2 + \log y^2 = \log z^2$, where x , y and z are positive numbers, which of the following must be true?

- I. $x^2 + y^2 = z^2$
- II. $\log x + \log y = \log z$
- III. $x^2 y^2 = z^2$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

Label 34 Let $F(x) = 2x^3 + 3x^2 - 11x - 6$. Given that $F(2) = 0$ and $F(-3) = 0$, then $F(x)$ can be factorized as

- A. $(x + 2)(x - 3)(2x + 1)$
- B. $(x + 2)(x - 3)(2x - 1)$
- C. $(x - 2)(x + 3)(2x + 1)$
- D. $(x - 2)(x - 3)(2x + 1)$
- E. $(x - 2)(x + 3)(2x - 1)$

Label 35 If a , b and c are positive numbers, which of the following is a possible graphical representation of $y = ax^2 + bx + c$

A

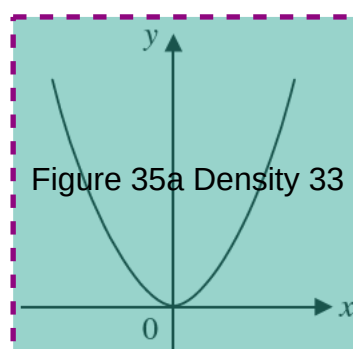


Figure 35a Density 33

B

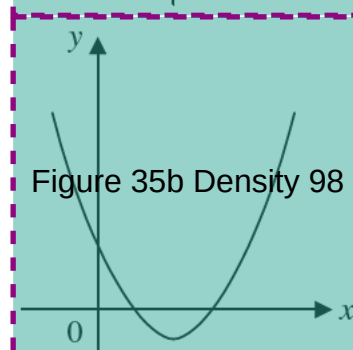


Figure 35b Density 98

C

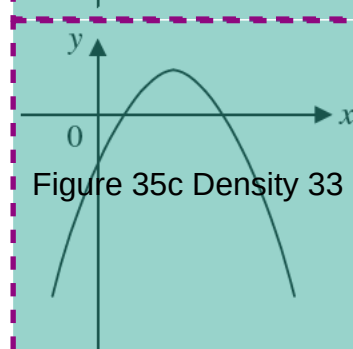


Figure 35c Density 33

D

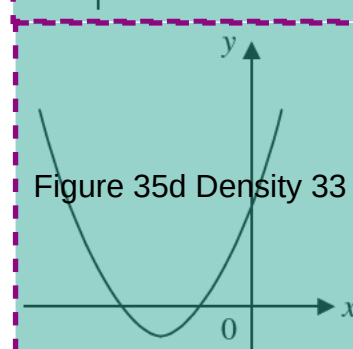


Figure 35d Density 33

E

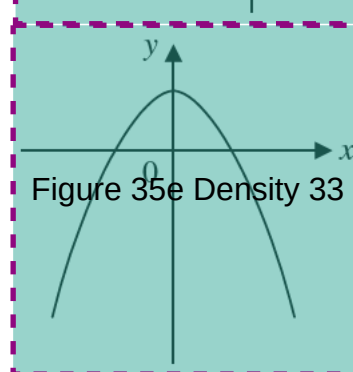


Figure 35e Density 33

Label 36
36. If $a > 0$ and $b < 0$, which of the following is/are negative?

- I. $\frac{1}{a} - \frac{1}{b}$
- II. $\frac{a}{b} + \frac{b}{a}$
- III. $\frac{a^2}{b} - \frac{b^2}{a}$

- A. I only
- B. III only
- C. I and II only
- D. I and III only
- E. II and III only

Label 37
37. If $2 < x < 3$ and $3 < y < 4$, then the range of values of $\frac{x}{y}$ is

- A. $\frac{1}{2} < \frac{x}{y} < \frac{3}{4}$
- B. $\frac{1}{2} < \frac{x}{y} < 1$
- C. $\frac{2}{3} < \frac{x}{y} < \frac{3}{4}$
- D. $\frac{2}{3} < \frac{x}{y} < 1$
- E. $\frac{4}{3} < \frac{x}{y} < \frac{3}{2}$

Label 38
38.

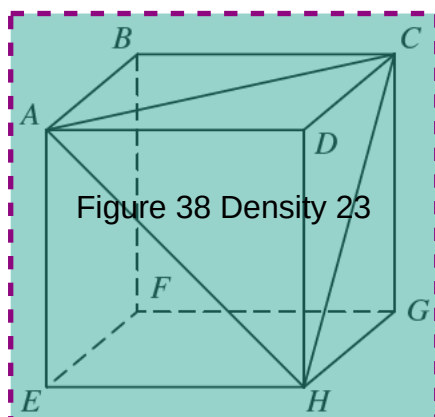


Figure 38 Density 23

$ABCDEFGH$ is a cube of side 3 cm. A tetrahedron $DACH$ is cut away along the plane ACH . The volume of the remaining solid is

- A. 6 cm^3
- B. 9 cm^3
- C. 13.5 cm^3
- D. 18 cm^3
- E. 22.5 cm^3

Label 39
39. The marked price of an article is originally P . The marked price is then increased so that when a discount of 10% is made on the new marked price, the selling price is still P . What is the new marked price?

- A. $\frac{9}{10}P$
- B. $\frac{109}{100}P$
- C. $\frac{11}{10}P$
- D. $\frac{111}{100}P$
- E. $\frac{10}{9}P$

Label 40
40.

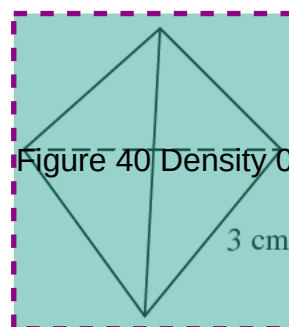


Figure 40 Density 0

The total surface area of a regular tetrahedron of side 3 cm is

- A. $\frac{9\sqrt{3}}{4} \text{ cm}^2$
- B. 9 cm^2
- C. $\frac{27\sqrt{3}}{4} \text{ cm}^2$
- D. $9\sqrt{3} \text{ cm}^2$

E. $12\sqrt{3} \text{ cm}^2$

E. Cannot be determined

Label 41
41. Ten litres of a mixture contain 60% of alcohol and 40% of water by volume. How many litres of water should be added so that it contains 30% of alcohol by volume?

- A. 5
- B. 10
- C. 15
- D. 20
- E. 60

Label 42
42. If A, B, and C can finish running the same distance in 3, 4 and 5 minutes respectively, then A's speed : B's speed : C's speed =

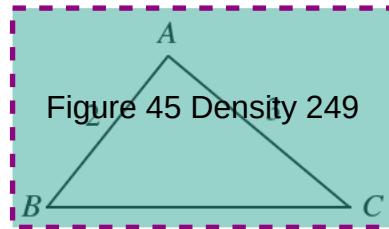
- A. 3 : 4 : 5
- B. 5 : 4 : 3
- C. 9 : 8 : 7
- D. 20 : 15 : 12
- E. 25 : 16 : 9

Label 43
43. If the five interior angles of a convex pentagon form an A.P. with a common difference of 10° , then the smallest interior angle of the pentagon is

- A. 52°
- B. 72°
- C. 88°
- D. 98°
- E. 108°

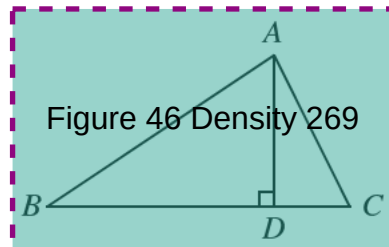
Label 44
44. Let p be a positive constant such that $p \sin \theta = \sqrt{3}$ and $p \cos \theta = 1$. Find all the values of θ in the interval 0 to 2π .

- A. $\frac{\pi}{3}$
- B. $\frac{\pi}{6}$
- C. $\frac{\pi}{3}, \frac{4\pi}{3}$
- D. $\frac{\pi}{6}, \frac{7\pi}{6}$



In $\triangle ABC$, $AB = 2$, $AC = 3$ and $\sin B = \frac{3}{4}$, then $\cos^2 C =$

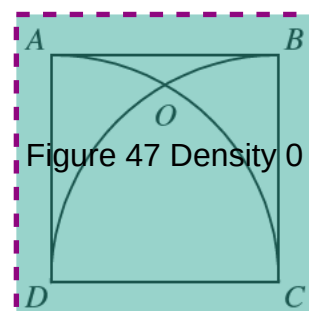
- A. $\frac{9}{16}$
- B. $\frac{9}{13}$
- C. $\frac{1}{4}$
- D. $\frac{1}{2}$
- E. $\frac{3}{4}$



In the figure, $BD : DC =$

- A. $\sin C : \sin B$
- B. $\cos C : \cos B$
- C. $\tan C : \tan B$
- D. $\sin B : \sin C$
- E. $\cos B : \cos C$

Label 47
47.



Label 49
49.
In the figure, $ABCD$ is a square. Arcs AC and BD are drawn with centres D and C respectively, intersecting at O .
Arc AQ : Arc OC =

- A. $1 : \sqrt{2}$
- B. $1 : \sqrt{3}$
- C. $1 : 2$
- D. $1 : 3$
- E. $2 : 3$

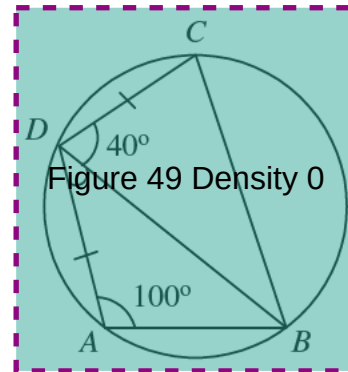


Figure 49 Density 0

DA and DC are equal chords of the circle $ABCD$. $\angle CDB = 40^\circ$ and $\angle DAB = 100^\circ$. $\angle ADB =$

- A. 20°
- B. 25°
- C. 30°
- D. 35°
- E. 40°

Label 48
48.

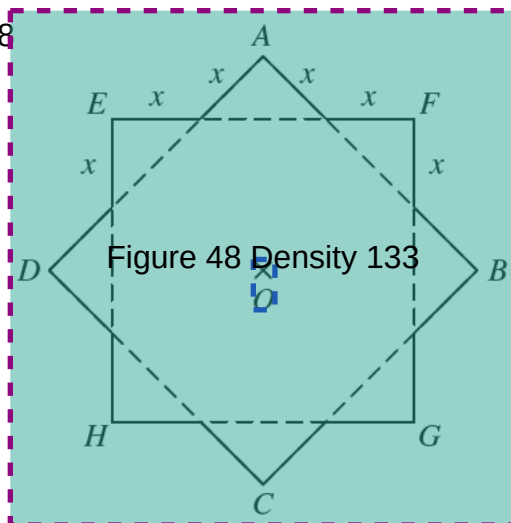


Figure 48 Density 133

In the figure, $ABCD$ and $EFGH$ are two squares of side 1. They are placed one upon the other with their centres both at O to form a star with 16 sides, each of length x . Find x .

- A. $\frac{2}{7}$
- B. $\frac{1}{3}$
- C. $\frac{2}{5}$
- D. $\frac{1}{2+\sqrt{2}}$
- E. $\frac{1}{1+\sqrt{2}}$

Label 50
50.

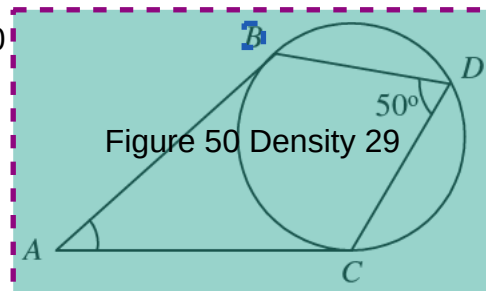


Figure 50 Density 29

In the figure, AB and AC are tangents to the circle BCD . If $\angle BDC = 50^\circ$, then $\angle A =$

- A. 130°
- B. 100°
- C. 85°
- D. 80°
- E. 60°

Label 51
51.

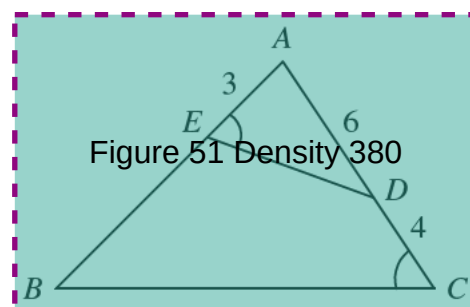
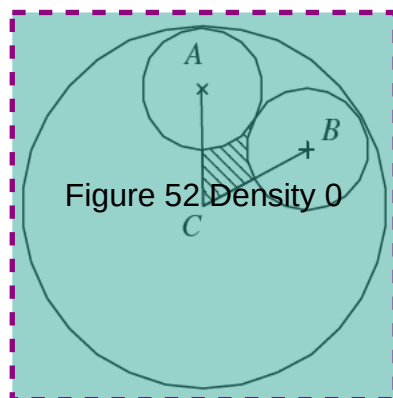


Figure 51 Density 380

In $\triangle ABC$, $AP = 3$, $AQ = 6$ and $QC = 4$.
If $\angle APQ = \angle ACB$, then $PB =$

- A. 7.
- B. 8.
- C. 10.
- D. 17.
- E. 20.

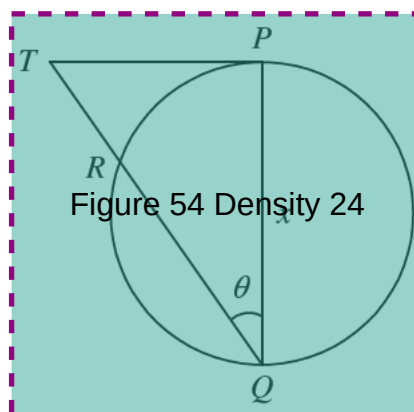
Label 52



A circle, centre O , touches the sector ABC internally at D , E and F . $\angle C = 60^\circ$ and $AC = 18$. Find the radius of the circle.

- A. 9.
- B. 6.
- C. 5.
- D. 4.
- E. 3.

Label 54



Three circles, centres A , B and C touch each other as shown in the figure. The radii of the two circles with centre A and B are both 1 cm and radius of the circle with centre C is 3 cm. Find the area of the shaded part in cm^2 .

- A. $\sqrt{3} - \frac{\pi}{3}$
- B. $\sqrt{3} - \frac{\pi}{6}$
- C. $2\sqrt{3} - \frac{\pi}{3}$
- D. $2\sqrt{3} - \frac{\pi}{6}$
- E. It cannot be determined

In the figure, PQ is a diameter and PT is a tangent of the circle. QT cuts the circle at R . Let $\angle Q = \theta$ and $PQ = x$, then $TR =$

- A. $\frac{x}{\cos \theta}$
- B. $\frac{x}{\sin \theta}$
- C. $\frac{x}{\sin \theta \tan \theta}$
- D. $x \sin \theta \tan \theta$
- E. $x \cos \theta \tan \theta$

Label 53

